

CHIDVILAS KUMKAPALLA

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PROFESSIONAL PROFILE

IT graduate with hands-on experience building and deploying production-ready AI systems using Python, AWS, and modern GenAI frameworks. Skilled in ML, data analytics, & MLOps pipelines with delivering real-world applications from RAG chatbots to sentiment analysis systems. Committed to solving impactful problems through scalable, data-driven solutions.

EDUCATION

DR BR Ambedkar National Institute of Technology Jalandhar	<i>2020-2024</i>
Information Technology	<i>CGPA: 7.32</i>
Narayana Junior College, Gudivada.	<i>2018-2020</i>
Higher Secondary – JEE Mains 2020: 99.29 percentile	<i>CGPA: 9.46</i>

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Power BI
- **Machine Learning & AI:** Machine Learning, GenAI, MLOps, Computer Vision, Scikit-learn
- **Systems & Cloud Technologies:** Data Structures, Cloud Computing
- **Tools & Technologies:** Hugging Face, Regex, REST APIs, RAG, Git, AWS, LangChain, Flask, Beautiful Soup, Prompt engineering.

WORK & RESEARCH EXPERIENCE

Cloud Computing Intern 📍 1Stop, Bengaluru

- Deployed and configured a scalable VM on Microsoft Azure to host a web application, ensuring 99.9% uptime during the internship period. Gained practical experience in cloud infrastructure management, security group configuration, and full-stack application deployment on AWS.

Data Science with GenAI Intern Nov 2025 – Present

- **MEDHA!** – Built a production-ready **Retrieval-Augmented Generation (RAG)** Q/A chatbot using Amazon Bedrock retrieve_and_generate API with Nova Pro LLM, OpenSearch vector store, and multi-turn conversational memory. Deployed on EC2 with UI, semantic document retrieval, & cited responses, reducing knowledge lookup time by eliminating manual doc search.
- **MindEase** – Developed and deployed a production-ready domain-specific AI chatbot using Google Gemini API, implementing advanced prompt engineering, multi-turn conversational memory, and modular architecture; deployed on AWS EC2 with secure environment variable configuration, public IP accessibility, and background process management.
- **Flipkart Review Sentiment Analysis (MLOps Pipeline)** Developing a sentiment classification pipeline for Flipkart product reviews. Implemented experiment tracking for multiple models (Logistic Regression, LinearSVC) while logging **parameters, metrics, and artifacts** with MLflow. Optimized model performance using **F1-score (0.96)** and registered the best-performing

model in the **MLflow Model Registry**. Integrated a **Prefect workflow** to orchestrate training tasks and demonstrate reproducible **MLOps experiment management**.

- **ReXplore** - Designed and deployed an webapp using Python, Flask & Jinja2, featuring regex operations, reducing manual regex debugging time by 80% for beginner-level learners.
- **IMGorithm** Developed a image processing app using Streamlit and OpenCV that enables real-time image transformations through UI. The app provides a dual-panel interface with a live preview and a scrollable control panel, allowing users to apply multiple image processing operations

PROJECTS

FarmSense360

Developed a platform to improve farming decisions with features such as a crop recommendation model for over 150 crops, crop price forecasting model using historical price data for over 30 different crops, an interactive community space, and diverse farming methods, leveraging various APIs to enrich the recommendation system. Crafted this system with React.js and Python in the backend, MongoDB database, and RESTful APIs, along with a UI built using Web-dev stack.

Clinical Readmission Prediction

Built and deployed **Clinical Readmission Predictor**, an ML application using **Python, Scikit-learn, and Streamlit**, trained on 25K hospital encounter records. Performed comprehensive **EDA** and feature-target correlation analysis. Designed a robust **ML pipeline** with categorical encoding and **feature scaling**. Evaluated best **5 models** (Random Forest, XGBoost, DT, KNN & Logistic Regression) and applied **Optuna-based hyperparameter optimization (200 trials)** with **Stratified K-Fold Cross-Validation**. Deployed the final model as an **interactive Streamlit Webapp on Hugging Face** enabling real-time readmission risk prediction.

Data Analysis Projects

- **Energy Consumption Analysis** – Engineered a data analytics project using **MySQL** to generate business insights from global energy consumption datasets and optimized complex SQL queries to identify trends across **200+ countries**.
- **Power BI Weather and Climate Dashboard** – Developed a **dynamic Power BI dashboard** using real-time datasets from the **Indian Meteorological Department (IMD)** and designed interactive visualizations with filters for year, region, and district.
- **CO₂ Emissions Data Analysis** – Analyzed large datasets (**50,000+ rows**) using **Python (Pandas, NumPy)** and **SQL** to clean, transform, and structure data, reducing missing values by **98%** through automated preprocessing. Performed statistical analysis (Z-score, correlation, skewness) and built visualization dashboards.
- **AutoEV: Web-Scraped Electric Vehicle Analytics** – Developed a **Python web scraper using BeautifulSoup** to extract specifications for **1150+ EV models**, cleaned data using **Regex**, and performed EDA with visualization and outlier detection.

KEY COMPETENCIES

Adaptable and quick learner with strong teamwork, leadership, and problem-solving abilities.

Extra-Curricular: Member, Team Cultural Affairs, NIT Jalandhar – Led and organized multiple cultural programs, fostering teamwork and inclusivity.

Certifications: NASSCOM FutureSkills Prime – Advanced Data Science with Python.